Technical Report: H3-Based Spatial Representation and Relationship Modeling for Heterogeneous Hierarchies

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Abstract

This report provides methodological transparency and reproducibility for the algorithms described in the associated research paper and thesis. It details the transformation of irregular spatial units into adaptive H3 hexagonal representations and the systematic computation of 17 qualitative spatial relationships.

1 Introduction

Traditional geographic information systems (GIS) predominantly represent place through geometric data models defined by precise spatial coordinates [1]. While excellent for discrete entities, these models often classify spatially separated geometries as simply `DISJOINT`, failing to capture graded proximity or multi-resolution interactions across hierarchical divisions [1]. This report presents a methodology using Uber's H3 Discrete Global Grid System (DGGS) as a computational bridge to link administrative, electoral, and postal hierarchies.

2 Algorithm 1: Adaptive H3 Cell Assignment

Spatial units (e.g., Welsh administrative, postal, and electoral divisions) form irregular tessellations that complicate consistent spatial analysis [1]. Algorithm 1 transforms these units into standardized hexagonal representations.

2.1 Mathematical Framework

- **Resolution Mapping Function** $T(A_i)$: Assigns an H3 resolution $r_i \in \{0, \ldots, 15\}$ based on the feature's area A_i in square meters [1].
- **Geometry Buffering (d_r)**: To ensure complete coverage of irregular boundaries, a buffer is applied in a meter-based CRS: $d_r = 0.5 \times \text{h3.edge_length}(r, \text{unit='m'})$ [1]. This ensures cell centroids near the boundary are included in the polyfill operation.
- **Fallback Mechanism**: To address small or irregular geometries where the initial resolution yields no cells, the resolution is incrementally increased: $r_i^{(k)} = T(A_i) + k$, for $k \in \{1, 2, 3\}$ [1].

2.2 Resolution Mapping Thresholds

Table 1 provides the complete thresholds used to assign resolutions. These ensure that assigned cells are substantially smaller than the spatial unit being represented to maintain geometric fidelity.

Table 1: Complete Adaptive H3 Resolution Mapping Thresholds $T(A_i)$

Resolution (r)	**Area Threshold** (A_i **in** m^2)	**Avg. Cell Area** (m^2)
0	$A_i \geq 609,788,441,794,100$	4,357,449,416,078.4
1	$86,801,780,399,000 \leq A_i < 609,788,441,794,100$	609,788,441,794.1
2	$12,393,434,655,100 \leq A_i < 86,801,780,399,000$	86,801,780,399.0
3	$1,000,000,000,000 \leq A_i < 12,393,434,655,100$	12,393,434,655.1
4	$100,000,000,000 \leq A_i < 1,000,000,000,000$	1,770,347,654.5
5	$10,000,000,000 \leq A_i < 100,000,000,000$	252,903,858.2
6	$1,000,000,000 \leq A_i < 10,000,000,000$	36,129,062.2
7	$100,000,000 \leq A_i < 1,000,000,000$	5,161,293.4
8	$10,000,000 \leq A_i < 100,000,000$	737,327.6
9	$1,000,000 \leq A_i < 10,000,000$	105,332.5
10	$500,000 \leq A_i < 1,000,000$	15,047.5
11	$100,000 \leq A_i < 500,000$	2,149.6
12	$10,000 \leq A_i < 100,000$	307.1
13	$1,000 \leq A_i < 10,000$	43.7
14	$100 \leq A_i < 1,000$	6.24
15	$A_i < 100$	0.89

3 Algorithm 2: Spatial Relationship Computation

Algorithm 2 identifies 17 relationships grouped into topological, proximity, and hierarchical categories [1].

3.1 Metric Logic

The overlap ratio ρ characterizes the degree of intersection:

$$\rho = \frac{|A \cap B|}{\min(|A|, |B|)}, \quad 0 \leq \rho \leq 1 \tag{1}$$

where A and B represent the H3 cell sets [1].

3.2 The 17 Spatial Relationships

Topological (Same Resolution): 1. **Identical**: Cell sets are equal. 2. **Complete Containment**: One set is a proper subset of another. 3. **Touch**: Boundary shared cells with $\rho \leq 0.05$. 4. **Intersect**: Shared cells with $\rho \leq 0.10$. 5. **Overlap**: Shared cells with $\rho > 0.10$. 6. **Disjoint**: No shared cells or adjacency.

Proximity (Same Resolution): Based on grid distance d: 7. **Neighbour**: $d = 1$. 8. **Close**: $d = 2$. 9. **Near**: $d \in \{3, 4\}$. 10. **Far**: $d \geq 5$. 11. **Far Away**: $d = -1$ (Undefined grid distance).

Hierarchical (Different Resolution): Uses parent-mapping (e.g., `cell_to_parent`): 12. **Direct Parent, Complete**: Finer unit fully geometrically contained by adjacent coarser resolution. 13. **Direct Parent, Partial**: Finer unit partially aligned to parent cells with boundary extension. 14. **Ancestor, Complete**: Full containment at non-adjacent resolution. 15. **Ancestor, Partial**: Partial containment at non-adjacent resolution. 16. **Hierarchical Touch**: Overlap limited to the boundary cells of the coarser unit. 17. **Hierarchical Overlap**: Overlap extending to the interior cells of the coarser unit [1].

4 Evaluation and Results

The framework was evaluated using 3,242 spatial units in Wales.

- **Speedup**: The H3 set-theoretic approach achieved a **3.6× speedup** over traditional vector DE-9IM computation (49 min vs 177 min) [1].
- **Fidelity**: Mean area approximation error was **8.30%** (SD 4.80%), confirming geometric fidelity suitable for qualitative reasoning [1].
- **Functional Equivalence**: The model identified **433 Identical** relationships compared to only 11 EQUALS in DE-9IM, successfully absorbing minor boundary discrepancies between heterogeneous datasets [1].

5 Data and Code Availability

The implementation of these algorithms is available at: `https://github.com/satoti/H3-Computation-Spati`

References

[1] Satoti, A. and Abdelmoty, A. I. (2026). H3-Based Computation of Qualitative Spatial Relationships Across Geographic Hierarchies. *Transactions in GIS / ISPRS International Journal of Geo-Information.*